One write, three readers

Every agent leaves a trail. Only your team can read it, and anyone can read the code.

THE PROBLEM

Teams run several coding agents, from Claude Code and Codex to open-source OpenCode and Goose, some on open-weight models they run themselves. Each harness keeps a partial record, and none controls how agents reach the world outside it.

Sessions vanish. Claude Code deletes them after 30 days.

Edits escape. Shell and subagent edits skip checkpoints.

Handoffs break. Work rarely follows you across harnesses.

Approvals go dark. Nothing records who clicked "Allow."

Reach is all or nothing. Cloud agents get everything, or none.

Local models leave no log. No vendor keeps one.

FOR DEVELOPERS

Never lose a session or an edit, whichever harness made it.

Every session goes to a vault under the developer's own keys, and every change gets a restore point, whatever tool path or model made it.

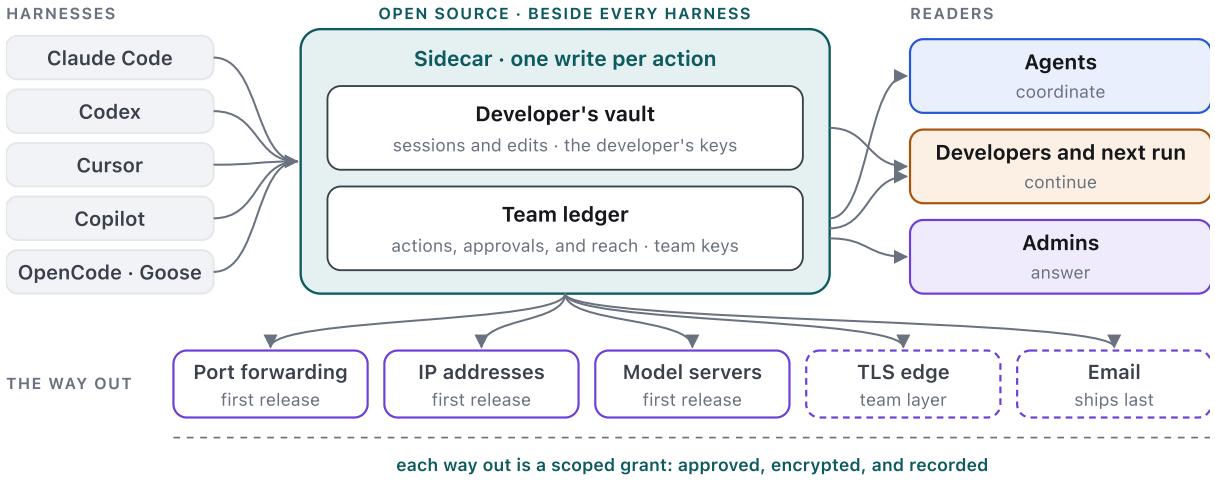
FOR SECURITY BUYERS

Every way your agents reach the outside world is private, approved, and on a record only your team can read.

Forwards, private addresses, and model servers first, then public names, then mail. Each is a scoped grant with a signed receipt.

HOW IT WORKS

Sidecar is a small open-source process beside any harness. It captures each action once, from the harness's hooks, the filesystem, and its own traffic, and writes it once: the raw session to the developer's vault, a signed entry to the team's ledger. Agents coordinate, developers continue, and admins answer from that one write.



WHY ONE WRITE MATTERS

Audit-only logs are overhead, so they get bypassed. Here agents need the entries to coordinate and developers need them to recover, so the record fills itself.

OPEN SOURCE

You don't have to trust us. The sidecar, relay, and entry format are open; self-host any part, air-gapped if needed. It is plain git underneath.

Private from us. Actions and approvals complete for your admins. Sessions stay with their author unless shared.

Never lose a session or an edit

Whichever harness made it. This is why a developer installs the sidecar.

~280
developers report agent changes they could not review or undo.

~200
report lost or hidden sessions. Claude Code deletes local transcripts after 30 days.
Claude Code #62476

139
report handoffs that break between runs, machines, or harnesses.

66%
of developers switched AI tools in the past year, so the record must outlive any one harness.
Stack Overflow survey 2026

The first three are counts of distinct people among 914 developer voices on GitHub, Hacker News, and forums, October 2026.

Session vault
WHY THEY INSTALL

Every session from every harness and model, under the developer's own keys. Archived instead of deleted, searchable on the developer's machine, secrets redacted before saving.

Change set and restore
WHY IT STAYS ON

Every file a run changed by any tool path, including shell commands, subagents, and reverts. Restore previews first, stays inside the run's own changes, and is recorded.

Pick up anywhere

Continue a run on another machine, in another harness or model, or as a teammate. Handoff notes record decisions, not transcripts.

Context that loads

Team context written into each harness's own files, such as `AGENTS.md` and `CLAUDE.md`, with a warning before a harness truncates it.
Codex silently cuts AGENTS.md at 32 KB.

Agents that find each other's work

Claims and leases across harnesses, so two agents don't edit the same files, and the next run starts where the last one stopped.
A hypothesis until the pilots measure it.

Yours unless you share

Admins see actions and approvals, not transcripts. Sessions open only when the developer shares them, and every admin read is visible to the developer.

A DAY WITHOUT LOSING ANYTHING

TIME	WHAT HAPPENS	WHAT THE SIDECAR KEEPS
09:10	A Claude Code session runs on a laptop.	The full session, in the developer's vault
10:20	A shell command edits files that the harness's checkpoints miss.	Every changed file, in the run's change set
11:05	The run goes wrong.	A preview, then a restore of only that run's changes
13:30	The developer continues in OpenCode, on a local model, on another machine.	The handoff note and the session, ready to resume
16:00	A teammate's agent starts on the same module.	A live claim, so that agent picks other work

Free for one person
Any number of agents and harnesses: the vault, restore, handoffs, and approval history.

Nothing to change
A subprocess any harness launches, over MCP on stdio, plus a CLI. No new editor.

Open source
Read exactly what it captures and redacts, or build it yourself.

Plain git underneath
Back up the vault, host it yourself, or leave with a clone.

Govern every way out

Every way your agents reach the outside world is private, approved, and on a record only your team can read.

Port forwarding

FIRST RELEASE

A cloud agent reaches a private service or a developer's own machine, without opening the network.

GRANT One agent, one port, one time window, approved by a person.

ENCRYPTED Between the two sidecars. The relay carries only ciphertext.

RECORDS `route` `approval` `ur bridge forward`

IP addresses

FIRST RELEASE

A permanent private address for each agent and service, plus subnets you choose to route.

GRANT Agents reach only granted ranges, and leave only through exits you run.

ENCRYPTED Packet by packet, between your sidecars.

RECORDS `route` `ur tun ip`

TLS edge

TEAM LAYER

A public HTTPS name for a service on a developer's machine, such as a client preview or a webhook.

GRANT Explicit, authenticated by default, and short-lived. Agent control panels are refused.

ENCRYPTED TLS passthrough. The certificate key stays in the sidecar.

RECORDS `publish` `ur bridge public-tls`

Email

SHIPS LAST

Addresses on your own domain for agents and people, including approval by reply.

GRANT Outside mail is untrusted. First messages to new recipients wait for approval.

ENCRYPTED End to end between members. Outside mail is encrypted on arrival.

RECORDS `ask` `answer` `approval` `ur id`

REACH IS ALL OR NOTHING TODAY

Codex cloud supports only Tailscale, and Tailscale warns that "By default, Codex could reach nearly everything in your tailnet." No harness vendor documents a cloud agent reaching a developer's own machine.

EASY EXPOSURE HAS ALREADY GONE WRONG

Knostic found 1,862 MCP servers open to the internet; none of 119 it checked required authentication to list tools. And 68% of organizations cannot tell agent activity from human activity (CSA, Aembit-sponsored).

WHAT ADMINS CAN ANSWER

- Which agents, harnesses, and models, down to the weights, touched which repos, for which sponsor.
- Who approved each action: one action ID, SSH-signed for high-risk steps.
- What was reachable, by whom, and who connected.
- Proof for a client that a person approved a change, without the code.
- One switch: revoking an agent cuts its forwards, addresses, names, and mail.

SAFE BY DEFAULT

The sidecar holds all three legs of the lethal trifecta: private data, untrusted input, and the way out. It keeps them apart.

- Untrusted mail cuts off send, forward, and publish until a person approves.
- Edits to `CLAUDE.md`, hooks, and `.mcp.json` go to review.
- Shared writes are scanned for secrets, and the sidecar keeps no long-lived ones.

WHY ENCRYPTION MAKES IT SAFE TO ADOPT

Adopting it adds no vendor that can read your agents' work: sessions stay under developers' keys, the ledger under the team's, and URnetwork holds only ciphertext, or nothing if you self-host. **Where it ends:** a hosted model sees what the harness sends, unless you run the model; outside mail arrives as ordinary SMTP; public names appear in certificate logs.

Open all the way down

Open source, so no one has to take our word for it. With open-weight models, nothing has to leave your control.

WHAT CHANGES WHEN THE MODEL IS YOURS

	HOSTED MODEL	OPEN WEIGHTS YOU RUN
Who sees prompts and code	The model provider	Only your team
Vendor record of agent work	Each vendor's, for its own product	None, so the record is the one you keep
What URnetwork holds	Ciphertext	Ciphertext, or nothing if you self-host
What the record can prove	Which harness and which model name	Also which weights, quantization, and runtime

Anthropic's Compliance API and OpenAI's compliance logs cover their own products. Open-weight teams get the whole privacy chain, and the most need for a record.

Model access

FIRST RELEASE

Agents reach the inference servers you run, such as vLLM, SGLang, Ollama, or llama.cpp, and nothing else.

GRANT Per agent and per model, approved by a person, revoked with the agent.

ENCRYPTED Between the sidecars. No vendor sits in the path.

PROVES The sidecar beside the model server signs which weights it serves.

RECORDS `route` `model` `ur bridge forward`

WHO IT'S FOR

Organizations that run open weights so no vendor reads their code

The best fit, since every part of the thesis holds. Also a minority: open-source models hold 11% of enterprise LLM spend, down from 19% a year earlier, in Menlo Ventures' estimate (an Anthropic investor).

Developers and homelabs

Adoption and credibility, but little revenue. Open tools draw big followings: OpenCode has 224k GitHub stars, Ollama 186k, llama.cpp 133k. Stars measure attention, not use.

WHERE OPEN TOOLS ALREADY STAND

PIECE	OPEN TOOLS TODAY	WHERE WE STAND
Reach	Free to self-host: frp (110k stars), Headscale (45k), zrok, and iroh. zrok's hosted service runs "the same code base" people self-host.	Free in ours too. We add per-agent grants and receipts.
Agents and shared models	Block's Buzz: people and agents in shared channels, and pooled spare compute, over iroh. More agent-to-agent is next.	Partner first. We record Goose like any harness.
Encrypted git	Radicle calls its private repos "unfit for use today"; git-crypt cannot revoke access; git-remote-gcrypt syncs only on push and pull.	Ours: real time, with team key rotation.
A signed record of agent work	Open parts: Tessera and Rekor v2 for logs, SCITT for receipts, an individual IETF draft for agent audit trails. Agent Receipts signs tool calls, but not sessions, edits, or reach.	Ours: sessions, edits, approvals, and reach, under your keys.

OPEN CORE, HOSTED BLIND

Free and open: the sidecar, admin app, relay, and entry format, and self-hosting any part, air-gapped if needed. **Paid:** relays, storage, and an edge we run and cannot read; names, certificates, and mail delivery; the team layer; support for your own installs. iroh, zrok, and Tailscale earn the same way.

Sidecar (working name) is in design. GitHub stars fetched 1 November 2026.

Sources: sidecar/research_notes, GitHub, Menlo Ventures

What's ours, and who pays

The tools are table stakes. The record under your keys is the product.

WHERE WE STAND, PIECE BY PIECE

PIECE	WHO ALREADY SELLS IT	WHERE WE STAND
Reach: forwards, private addresses, public names	ngrok, Cloudflare Tunnel, and Tailscale, which is end-to-end encrypted and Codex cloud's only VPN	At par, and free when self-hosted. We add per-agent grants and receipts.
Agent mail	AgentMail, Resend Inboxes, Cloudflare Email Service	At par, so it ships last.
Agent memory and storage	Cloudflare Artifacts, Letta MemFS, Entire, claude-mem	At par, plus keys the customer holds.
Coordination across agents	Claude Code agent teams, Codex message boards, and Block's Buzz: one vendor each	Ahead across harnesses. The value is still to be measured.
A record of agent work you can read and we cannot	No one. Harness vendors bind it to their runtime; gateways and recorders read it.	Ours.
Sessions and actions under separate keys	Span keeps sessions private by access control, not keys.	Ours.

Why the gap stays open: harness owners earn when work stays on their runtime, and gateway and recorder vendors are built on reading the data. That is an inference about incentives. Tailscale's and Cloudflare's roadmaps are the first to watch.

WHO PAYS

Solo **Free, open source**	One person, any number of agents: the vault, restore, handoffs, forwards, and model access.
Team **$8–25 / seat***	From the second person, hosted: the shared signed ledger, inventory, reach receipts, client packets, SSO, exports.
Assured **~$39 / seat***	SCIM, one kill switch, a visible audit member, legal hold, compliance mappings; on your servers too.

*Inferred from Tailscale ($8), Agent 365 ($15), Copilot Business ($19), and Claude Team ($20–25). Agents are never metered; hosted storage, edge traffic, and mail are. Self-hosting is free.

WHERE WE START

Teams of 5 to 150 engineers on two or more harnesses, where proof is already demanded:

- Consultancies and agencies under client AI-disclosure terms
- Microsoft suppliers, whose change logs must name the approver
- Teams whose cloud agents need private reach
- Fintech, health, and other regulated teams, including those on open weights
- Agent vendors facing AIUC-1 reviews

MONEY IS MOVING INTO THIS SPACE

- **Entire:** $60M seed (Feb 2026), agent sessions in git
- **AgentMail:** $6M seed (Mar 2026), agent email
- **Tailscale:** $160M Series C (Apr 2025), now for agents
- **Span:** $25M (Nov 2025), agent traces
- **Keycard** ($38M), **Mem0** ($24M): identity, memory
- **Agent 365:** $15 per user, agents unlicensed

Now
Questionnaires ask which AI a team uses and whether a human reviews its work.

December 2027
Draft SOC 2 criteria, EU AI Act high-risk logging, and most Cyber Resilience Act duties apply.

The moat is time and service
A team that starts in 2026 holds a year of signed approvals when audits arrive. The format is open, so teams stay because the service is good.

Proof before scale

One open-source capture engine, ten to fifteen design partners, and published trust documents come before the rest of the build.

WHAT SHIPS, IN ORDER

First release, open source

The vault, change sets and restore, approval receipts, run inventory and agent identity, handoffs and claims, commit-to-run links, and the instruction-file writer.

Reach: port forwarding, IP addresses, and model access.

Next: the team layer

The shared signed ledger, SSO and SCIM, SIEM export, retention, the verifiable log, client packets, reach receipts, and a supported self-hosted install.

Reach: the TLS edge, and approval by email reply.

Last: inbox and assurance

The visible audit member, key-escrow options, legal hold, compliance mappings, and an air-gapped install.

Reach: mail with the outside world on your domain, and group inboxes.

Each stage ships whole. Reach comes first, as the closest to code URnetwork runs today; email, the heaviest build, comes last.

TEN TO FIFTEEN DESIGN PARTNERS

Teams of 5 to 150 engineers on two or more harnesses, drawn from the beachhead, including two or three that run open weights. Each question has a measure, and a decision if the answer is no.

QUESTION	WHAT WE MEASURE	IF THE ANSWER IS NO
Do developers keep it running?	Daily use of the vault and restore, per developer	Rework the developer hook before building the team layer
Does the ledger sell?	Conversion to Team when a second person joins	Move the free line or the Team price
Does coordination earn its place?	Duplicate work and conflicting edits, with and without claims	Keep claims as a free feature, not a selling point
Is reach a wedge?	Cloud-agent grants approved and used each week	Keep reach as a feature, and lead with the record
Will teams pay?	Paid conversions at $8–25 a seat	Re-price before launch
Do open-weight teams adopt it, and pay?	Model calls routed through the sidecar; hosted versus self-hosted relays	Sell the team layer and support, not hosting

TRUST DOCUMENTS FIRST

Before "end-to-end encrypted" appears in public:

- The sidecar, admin app, relay, and entry format ship open source, with reproducible builds.
- An independent audit covers the client and the protocol.
- A threat model assumes a malicious server and lists all metadata our servers see.
- A dated warrant canary and current transparency reports return.
- A plain statement: the sidecar has no token and uses URnetwork's network only for worldwide reach.

RISKS WE MANAGE

- **Scope.** One capture engine and one entry format; each stage ships whole.
- **Completeness.** Hooks fail open, so each run states what the record could not see.
- **Free built-ins and self-hosting.** We sell hosting we can't read, the team layer, and support.
- **Open rivals.** Block's Buzz is closest; we partner first and record Goose like any harness.
- **Reputation.** Trust documents come before claims, and agents leave only through exits you run.